SUTEX BANK COLLEGE OF COMPUTER APPLICATIONS & SCIENCE, AMROLI
MSc IT PROGRAMME
MODEL QUESTION PAPER - 1

Course: FY MSc IT / BSc IT	**Year: 2026-27**
Semester: 1st	**Total Marks : 50**
Subject: 104 Fundamentals of Programming Using C-1	**Time: 2 Hours**

Q.1	**Do as directed: (Attempt any Ten)**	**[10]**
1	Who developed the C programming language and in which year?	
2	Which format specifier is used to read or print an integer value in C? (A) %c (B) %d (C) %f (D) %s	
3	What is the output of the arithmetic expression 12 + 4 * 2 in C?	
4	In a flowchart, which geometric symbol is used for Start and Stop (Terminal)?	
5	Which keyword is used to terminate or exit from a loop prematurely? (A) exit (B) return (C) break (D) continue	
6	True or False: A variable name in C can start with a digit.	
7	What will be the output of printf("%d", 15 > 8); in C?	
8	An assembler translates which language into machine code? (A) High-level language (B) Assembly language (C) Natural language (D) Binary code	
9	What is the size in bytes occupied by a float data type in standard C?	
10	Which operator is used for Logical AND in C? (A) & (B) && (C) | (D) ||	

11	What will be the output of the following code snippet?
	int x = 5;
	printf("%d", ++x);	

12	State the meaning and syntax of the sizeof operator in C.	
Q.2 (A)	**Answer following questions: (Attempt any Two)**	**[10]**
(1) (2) (3)	What is a Constant in C? Explain Integer, Floating-point, Character, and String constants with examples. Explain the concept of Divide and Conquer technique in problem solving with a suitable everyday example. Explain the switch...case statement with syntax, rules, and a simple program demonstrating a menu choice.	
Q.2 (B)	**Answer following question: (Attempt any One)**	**[04]**

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(1) **OR** (2)	Differentiate between pre-increment (++x) and post-increment (x++) with simple illustrative code. Explain the difference between Compiler and Interpreter with neat points of comparison.	
Q.3 (A)	**Answer following questions: (Attempt any Two)**	**[10]**
(1) (2) (3)	What is a Flowchart? Explain the five standard flowchart symbols with neat sketches and their functions. Explain the structure of a C program in detail with an annotated example showing preprocessor directives and main(). What is looping? Explain for loop with its syntax, flowchart, and a program to print numbers from 1 to 10.	
Q.3 (B)	**Answer following question: (Attempt any One)**	**[04]**
(1) **OR** (2)	Explain the Conditional (Ternary) operator (?:) in C with syntax and an example to find larger of two numbers. Differentiate between Syntax errors and Logical errors with simple examples of each.	
Q.4	**Answer following questions: (Attempt any Three)**	**[12]**
(1) (2) (3) (4)	Write a C program to check whether a given year entered by the user is a Leap Year or not. Draw a flowchart and write an algorithm to find the Simple Interest given Principal, Rate, and Time. Write a C program to print the multiplication table of a given integer N (e.g., N x 1 to N x 10). Write a C program to find the reverse of a given positive integer (e.g., if input is 1234, output is 4321).	